UNIVERSITY PARTNER

UNIVERSITY OF WOLVERHAMPTON

HERALD COLLEGE KATHMANDU

6CS012 Artificial Intelligence and Machine Learning

Project Proposal

Flower Image Classification

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Group: L6CG13

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1 Project Proposal

1.1 Group Members:

Manjushree Tamang, Prinsha Shrestha, Priya Shrestha, Riddhi Maskey

1.2 Research Questions:

Can a deep learning-based image classification model accurately identify different species of flowers based on their visual characteristics?

1.3 Title

Flower Species Classification

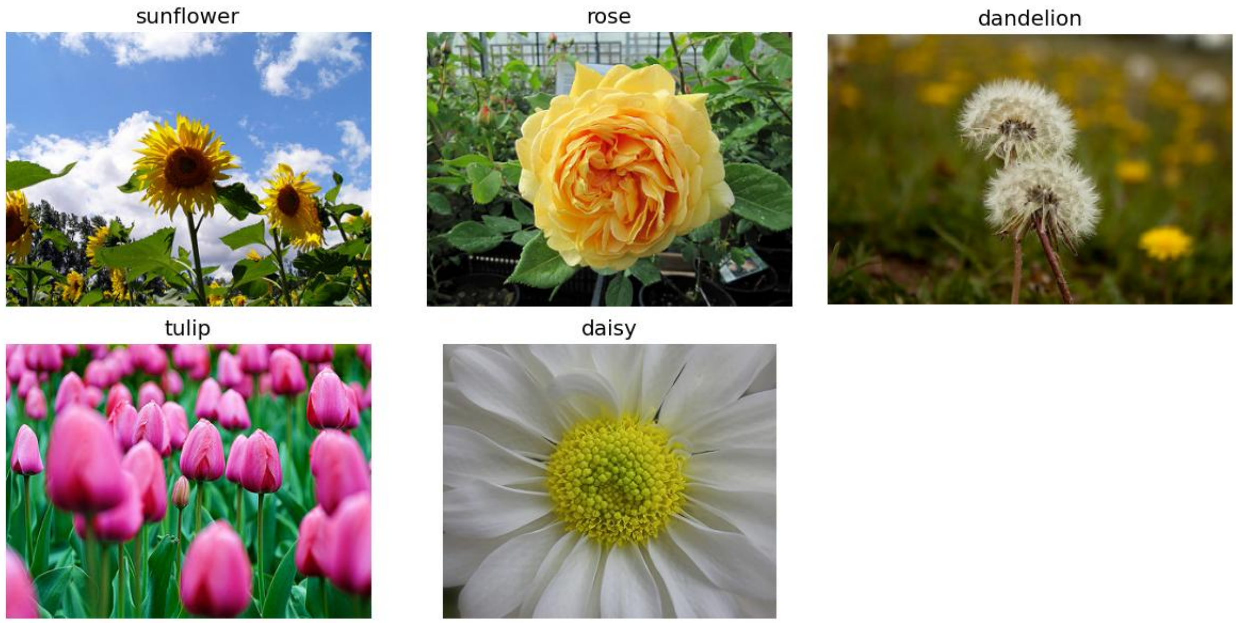
1.4 Dataset Description

- **Dataset:** Flower Classification
- **Source:** This dataset was chosen from the recommended datasets provided in the shared drive folder.
- **Number of Images:** The dataset consists of a total of 4340 images.
  - ❖ Daisy
  - ❖ Tulip
  - ❖ Dandelion
  - ❖ Rose
  - ❖ Sunflower
- **Image Distribution:** Each class has the following image count.

```
Total Images: 4340
Class Distribution:
       Class  Image Count
4  dandelion         1061
2      tulip          991
0       rose          783
3      daisy          763
1  sunflower          742
```

1.5 Data Visualization:

Random flower images from each class:



Class distribution visualization